

BridgeX Future Developer Operations & Scale Manual

A 600-page living reference for architecture, workflow safety, one-million-user growth, operations, and troubleshooting

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1 Volume I — Operating Boundary and Product Model

This volume establishes the safety boundary: BridgeX provides public discovery and protected execution. It does not make unsupported banking, customs, airline, legal, or identity claims. The source of truth for access is authenticated database policy and protected backend logic, not a client-side visual state.

Manual use and change-control contract

How a new developer should use this publication

Purpose. Use this manual as an operating map, then confirm exact current behavior in source code, migration history, and production configuration before making a change.

Primary location	docs/BridgeX_Future_Operations_Manual/
Inputs / prerequisites	Repository clone, least-privilege access, and a safe test environment
Expected outputs	A documented, reviewable implementation plan
Security or reliability boundary	Never treat screenshots, UI labels, or client code as the sole authority for privacy or payment behavior.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A peer can follow the plan, locate the source of truth, and reproduce the validation steps.

Scaling note. At larger scale, distribute ownership by domain and retain an explicit service owner for every workflow.

Role and permission boundary

Guest, member, administrator, and super-admin roles

Purpose. Separate public discovery from protected execution and prevent privilege from being inferred from a visible screen.

Primary location	Supabase RLS, role checks, AdminControl and SuperAdminControl routes
Inputs / prerequisites	Authenticated identity, role value, and scoped record identifier
Expected outputs	Only the minimum permitted fields and actions
Security or reliability boundary	Passwords, private documents, payment proofs, private addresses, and unrestricted chat are not public and are not recoverable from UI controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run role-based tests for guest, member, administrator, and super-admin paths.

Scaling note. At one million users, authorization must remain database-enforced; never replace RLS with a front-end-only role test.

One-million-user planning baseline

Capacity planning before high-concurrency growth

Purpose. Scale based on peak concurrent reads, writes, uploads, realtime connections, and payment review volume rather than registered account count.

Primary location	docs/BridgeX_Developer_Operations_Handbook.md and future infrastructure configuration
Inputs / prerequisites	Measured active users, request rate, database latency, error rate, queue depth, and storage growth
Expected outputs	Capacity decision, budget forecast, and explicit scale trigger
Security or reliability boundary	Do not extrapolate from a single free instance or rely on a platform-wide availability assumption.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Load-test a staging environment and capture p95 latency, error rate, database CPU, and queue depth.

Scaling note. Use staged load testing and admission controls before launch events; separate cacheable public traffic from protected traffic.

1.1 Core architecture diagrams

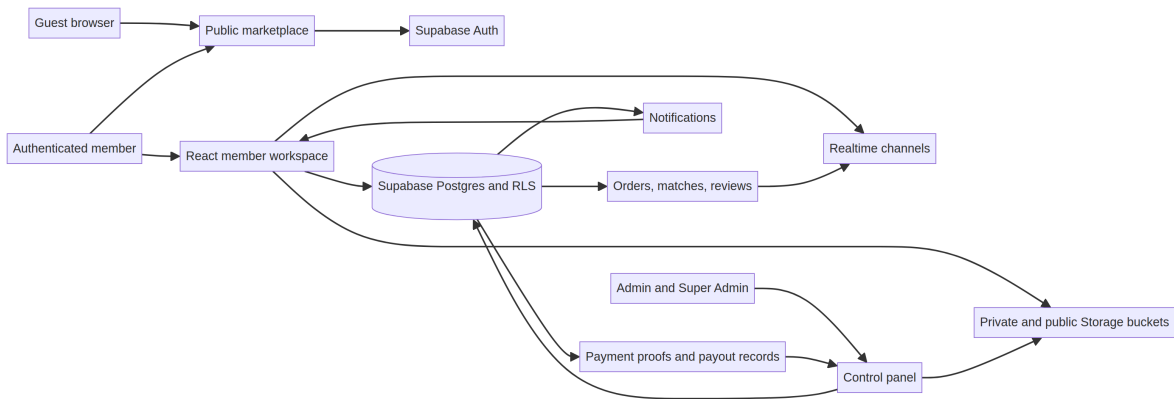


图 1 Figure: BridgeX public discovery, protected workspace, database controls, storage, and administrator operations.

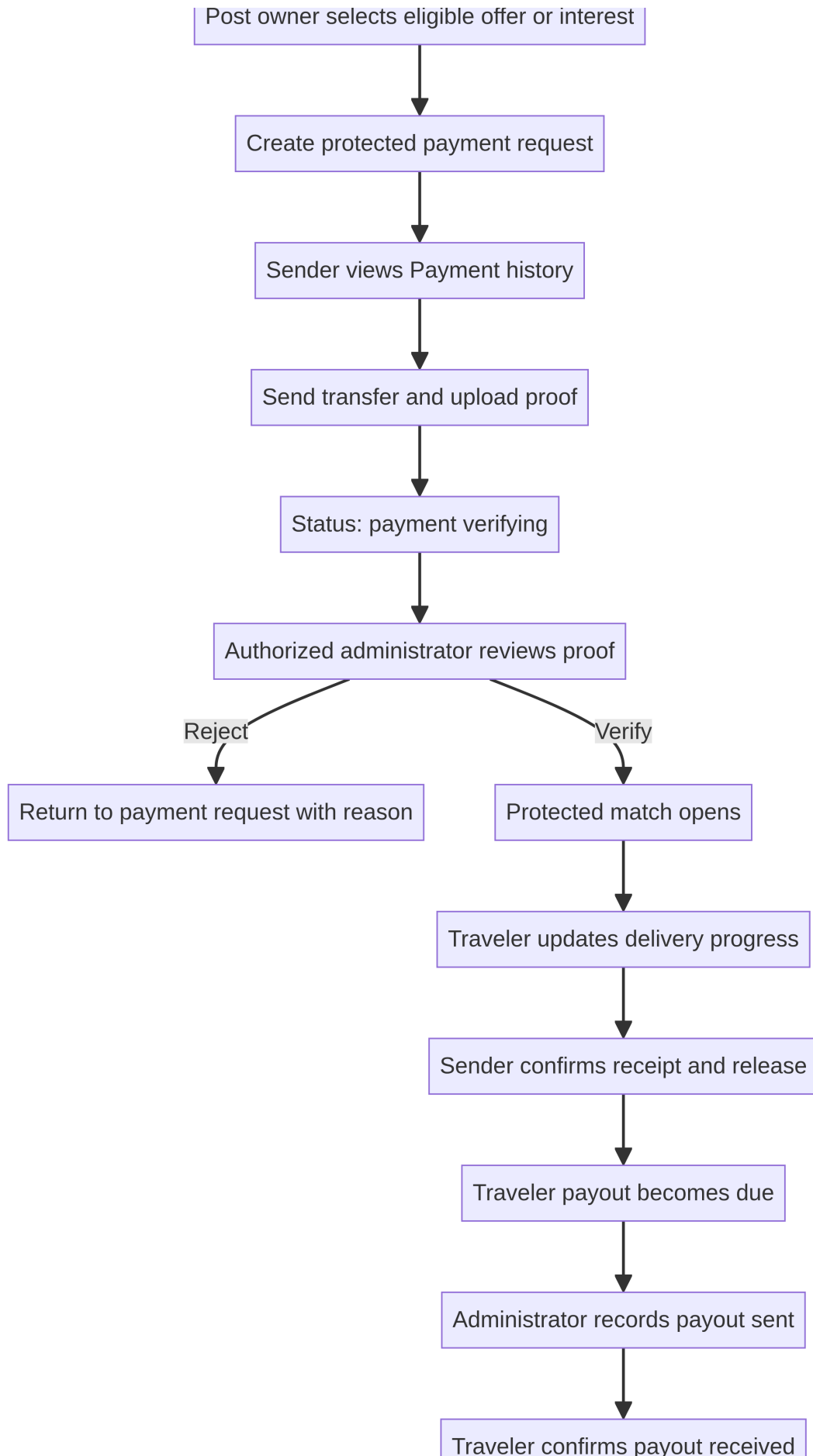


图 2 Figure: protected payment proof, match opening, delivery release, and traveler payout lifecycle.

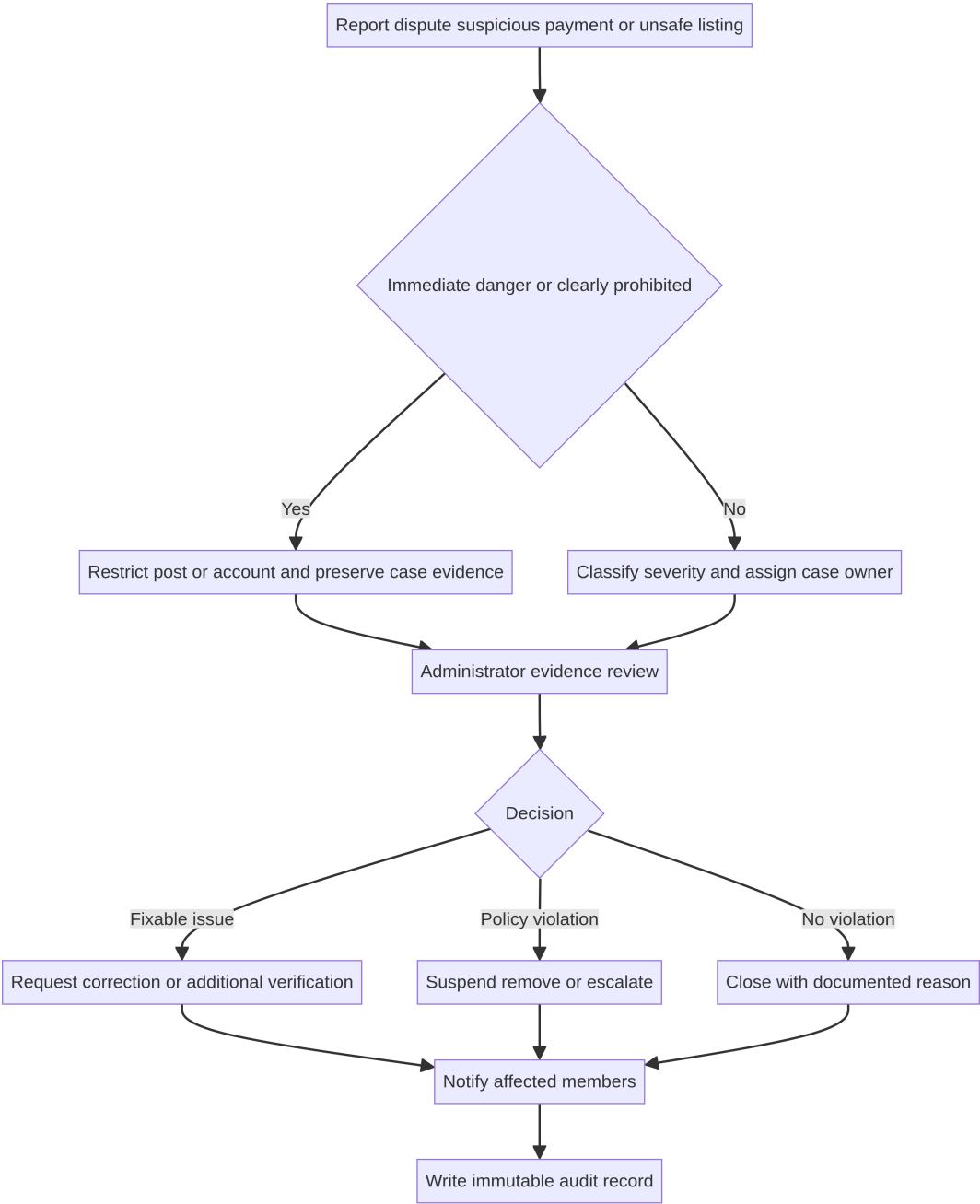


图 3 Figure: incident classification, evidence preservation, decision, notification, and audit record.

2 Volume II — Real Repository and Interface Map

Every generated sheet in this volume points to a real source file. It records safe modification, diagnostic, and validation expectations without claiming that a filename alone proves an authorization boundary.

Source map — App.tsx

client implementation module

Purpose. This sheet documents the real repository unit `apps/web/client/src/App.tsx` and its visible exported or local symbols: Home, Marketplace, Access, ResetPassword, Workspace.

Primary location	<code>apps/web/client/src/App.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Use a narrow change and trace all imports before refactoring.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — useAuth.ts

shared client utility

Purpose. This sheet documents the real repository unit `apps/web/client/src/_core/hooks/useAuth.ts` and its visible exported or local symbols: `SESSION_CACHE_MS`, `readBridgeXUser`, `authenticatedUser`, `metadata`, `avatarPath`.

Primary location	<code>apps/web/client/src/_core/hooks/useAuth.ts</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Avoid creating unbounded browser work or storing private data in browser persistence.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — AIChatBox.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/AIChatBox.tsx` and its visible exported or local symbols: `ChatPage`, `chatMutation`, `handleSend`, `newMessages`, `AIChatBox`.

Primary location	<code>apps/web/client/src/components/AIChatBox.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — DashboardLayout.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/DashboardLayout.tsx` and its visible exported or local symbols: `memberMenu`, `adminMenu`, `DashboadLayout`, `previewUser`, `currentUser`.

Primary location	<code>apps/web/client/src/components/DashboardLayout.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — DashboardLayoutSkeleton.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/DashboardLayoutSkeleton.tsx` and its visible exported or local symbols: `DashboardLayoutSkeleton`.

Primary location	<code>apps/web/client/src/components/DashboardLayoutSkeleton.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — ErrorBoundary.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ErrorBoundary.tsx` and its visible exported or local symbols: `staleModule`, `reloadKey`, `url`.

Primary location	<code>apps/web/client/src/components/ErrorBoundary.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — ManusDialog.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ManusDialog.tsx` and its visible exported or local symbols: `ManusDialog`, `handleOpenChange`.

Primary location	<code>apps/web/client/src/components/ManusDialog.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — Map.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/Map.tsx` and its visible exported or local symbols: `mapRef`, `place`, `geocoder`, `dist`, `directionsService`.

Primary location	<code>apps/web/client/src/components/Map.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — Brand.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/bridgex/Brand.tsx` and its visible exported or local symbols: `BRIDGEX_LOGO_URL`, `Brand`.

Primary location	<code>apps/web/client/src/components/bridgex/Brand.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — DeliveryLoader.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/bridgex/DeliveryLoader.tsx` and its visible exported or local symbols: `DeliveryLoader`.

Primary location	<code>apps/web/client/src/components/bridgex/DeliveryLoader.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — PublicLayout.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/bridgex/PublicLayout.tsx` and its visible exported or local symbols: `navigation`, `displayName`, `ANDROID_BUILD`, `ANDROID_DOWNLOAD_URL`, `AndroidUpdatePrompt`.

Primary location	<code>apps/web/client/src/components/bridgex/PublicLayout.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — VerifiedBadge.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/bridgex/VerifiedBadge.tsx` and its visible exported or local symbols: `VerifiedBadge`, `MemberVerificationBadge`.

Primary location	<code>apps/web/client/src/components/bridgex/VerifiedBadge.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — accordion.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/accordion.tsx` and its visible exported or local symbols: `Accordion`, `AccordionItem`, `AccordionTrigger`, `AccordionContent`.

Primary location	<code>apps/web/client/src/components/ui/accordion.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — alert-dialog.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/alert-dialog.tsx` and its visible exported or local symbols: `AlertDialog`, `AlertDialogTrigger`, `AlertDialogPortal`, `AlertDialogOverlay`, `AlertDialogContent`.

Primary location	<code>apps/web/client/src/components/ui/alert-dialog.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — alert.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/alert.tsx` and its visible exported or local symbols: `alertVariants`, `Alert`, `AlertTitle`, `AlertDescription`.

Primary location	<code>apps/web/client/src/components/ui/alert.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — aspect-ratio.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/aspect-ratio.tsx` and its visible exported or local symbols: `AspectRatio`.

Primary location	<code>apps/web/client/src/components/ui/aspect-ratio.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — avatar.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/avatar.tsx` and its visible exported or local symbols: `Avatar`, `AvatarImage`, `AvatarFallback`.

Primary location	<code>apps/web/client/src/components/ui/avatar.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — badge.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/badge.tsx` and its visible exported or local symbols: `badgeVariants`, `Badge`, `Comp`.

Primary location	<code>apps/web/client/src/components/ui/badge.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — breadcrumb.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/breadcrumb.tsx` and its visible exported or local symbols: `Breadcrumb`, `BreadcrumbList`, `BreadcrumbItem`, `BreadcrumbLink`, `Comp`.

Primary location	<code>apps/web/client/src/components/ui/breadcrumb.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — button-group.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/button-group.tsx` and its visible exported or local symbols: `buttonGroupVariants`, `ButtonGroup`, `ButtonGroupText`, `Comp`, `ButtonGroupSeparator`.

Primary location	<code>apps/web/client/src/components/ui/button-group.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — button.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/button.tsx` and its visible exported or local symbols: `buttonVariants`, `Button`, `Comp`.

Primary location	<code>apps/web/client/src/components/ui/button.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — calendar.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/calendar.tsx` and its visible exported or local symbols: `Calendar`, `defaultClassNames`, `CalendarDayButton`, `ref`.

Primary location	<code>apps/web/client/src/components/ui/calendar.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — card.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/card.tsx` and its visible exported or local symbols: `Card`, `CardHeader`, `CardTitle`, `CardDescription`, `CardAction`.

Primary location	<code>apps/web/client/src/components/ui/card.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — carousel.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/carousel.tsx` and its visible exported or local symbols: `CarouselContext`, `useCarousel`, `context`, `Carousel`, `onSelect`.

Primary location	<code>apps/web/client/src/components/ui/carousel.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — chart.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/chart.tsx` and its visible exported or local symbols: `THEMES`, `ChartContext`, `useChart`, `context`, `ChartContainer`.

Primary location	<code>apps/web/client/src/components/ui/chart.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — checkbox.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/checkbox.tsx` and its visible exported or local symbols: `Checkbox`.

Primary location	<code>apps/web/client/src/components/ui/checkbox.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — collapsible.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/collapsible.tsx` and its visible exported or local symbols: `Collapsible`, `CollapsibleTrigger`, `CollapsibleContent`.

Primary location	<code>apps/web/client/src/components/ui/collapsible.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — command.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/command.tsx` and its visible exported or local symbols: `Command`, `CommandDialog`, `CommandInput`, `CommandList`, `CommandEmpty`.

Primary location	<code>apps/web/client/src/components/ui/command.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — context-menu.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/context-menu.tsx` and its visible exported or local symbols: `ContextMenu`, `ContextMenuTrigger`, `ContextMenuGroup`, `ContextMenuPortal`, `ContextMenuSub`.

Primary location	<code>apps/web/client/src/components/ui/context-menu.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — dialog.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/dialog.tsx` and its visible exported or local symbols: `DialogCompositionContext`, `useDialogComposition`, `Dialog`, `composingRef`, `justEndedRef`.

Primary location	<code>apps/web/client/src/components/ui/dialog.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — drawer.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/drawer.tsx` and its visible exported or local symbols: `Drawer`, `DrawerTrigger`, `DrawerPortal`, `DrawerClose`, `DrawerOverlay`.

Primary location	<code>apps/web/client/src/components/ui/drawer.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — dropdown-menu.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/dropdown-menu.tsx` and its visible exported or local symbols: `DropdownMenu`, `DropdownMenuPortal`, `DropdownMenuTrigger`, `DropdownMenuContent`, `DropdownMenuGroup`.

Primary location	<code>apps/web/client/src/components/ui/dropdown-menu.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — empty.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/empty.tsx` and its visible exported or local symbols: `Empty`, `EmptyHeader`, `emptyMediaVariants`, `EmptyMedia`, `EmptyTitle`.

Primary location	<code>apps/web/client/src/components/ui/empty.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — field.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/field.tsx` and its visible exported or local symbols: `FieldSet`, `FieldLegend`, `FieldGroup`, `fieldVariants`, `Field`.

Primary location	<code>apps/web/client/src/components/ui/field.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — form.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/form.tsx` and its visible exported or local symbols: `Form`, `FormFieldContext`, `FormField`, `useFormField`, `fieldContext`.

Primary location	<code>apps/web/client/src/components/ui/form.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — hover-card.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/hover-card.tsx` and its visible exported or local symbols: `HoverCard`, `HoverCardTrigger`, `HoverCardContent`.

Primary location	<code>apps/web/client/src/components/ui/hover-card.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — input-group.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/input-group.tsx` and its visible exported or local symbols: `InputGroup`, `inputGroupAddonVariants`, `InputGroupAddon`, `inputGroupButtonVariants`, `InputGroupButton`.

Primary location	<code>apps/web/client/src/components/ui/input-group.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — input-otp.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/input-otp.tsx` and its visible exported or local symbols: `InputOTP`, `InputOTPGroup`, `InputOTPSlot`, `inputOTPContext`, `InputOTPSeparator`.

Primary location	<code>apps/web/client/src/components/ui/input-otp.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — input.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/input.tsx` and its visible exported or local symbols: `Input`, `dialogComposition`, `isComposing`.

Primary location	<code>apps/web/client/src/components/ui/input.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — item.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/item.tsx` and its visible exported or local symbols: `ItemGroup`, `ItemSeparator`, `itemVariants`, `Item`, `Comp`.

Primary location	<code>apps/web/client/src/components/ui/item.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — kbd.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/kbd.tsx` and its visible exported or local symbols: `Kbd`, `KbdGroup`.

Primary location	<code>apps/web/client/src/components/ui/kbd.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — label.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/label.tsx` and its visible exported or local symbols: `Label`.

Primary location	<code>apps/web/client/src/components/ui/label.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — menubar.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/menubar.tsx` and its visible exported or local symbols: `Menubar`, `MenubarMenu`, `MenubarGroup`, `MenubarPortal`, `MenubarRadioGroup`.

Primary location	<code>apps/web/client/src/components/ui/menubar.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — navigation-menu.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/navigation-menu.tsx` and its visible exported or local symbols: `NavigationMenu`, `NavigationMenuList`, `NavigationMenuItem`, `navigationMenuTriggerStyle`, `NavigationMenuTrigger`.

Primary location	<code>apps/web/client/src/components/ui/navigation-menu.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — pagination.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/pagination.tsx` and its visible exported or local symbols: `Pagination`, `PaginationContent`, `PaginationItem`, `PaginationLink`, `PaginationPrevious`.

Primary location	<code>apps/web/client/src/components/ui/pagination.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — popover.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/popover.tsx` and its visible exported or local symbols: `Popover`, `PopoverTrigger`, `PopoverContent`, `PopoverAnchor`.

Primary location	<code>apps/web/client/src/components/ui/popover.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — progress.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/progress.tsx` and its visible exported or local symbols: `Progress`.

Primary location	<code>apps/web/client/src/components/ui/progress.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — radio-group.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/radio-group.tsx` and its visible exported or local symbols: `RadioGroup`, `RadioGroupItem`.

Primary location	<code>apps/web/client/src/components/ui/radio-group.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — resizable.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/resizable.tsx` and its visible exported or local symbols: `ResizablePanelGroup`, `ResizablePanel`, `ResizableHandle`.

Primary location	<code>apps/web/client/src/components/ui/resizable.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — scroll-area.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/scroll-area.tsx` and its visible exported or local symbols: `ScrollArea`, `ScrollBar`.

Primary location	<code>apps/web/client/src/components/ui/scroll-area.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — select.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/select.tsx` and its visible exported or local symbols: `Select`, `SelectGroup`, `SelectValue`, `SelectTrigger`, `SelectContent`.

Primary location	<code>apps/web/client/src/components/ui/select.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — separator.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/separator.tsx` and its visible exported or local symbols: `Separator`.

Primary location	<code>apps/web/client/src/components/ui/separator.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — sheet.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/sheet.tsx` and its visible exported or local symbols: `Sheet`, `SheetTrigger`, `SheetClose`, `SheetPortal`, `SheetOverlay`.

Primary location	<code>apps/web/client/src/components/ui/sheet.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — sidebar.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/sidebar.tsx` and its visible exported or local symbols: `SIDEBAR_COOKIE_NAME`, `SIDEBAR_COOKIE_MAX_AGE`, `SIDEBAR_WIDTH`, `SIDEBAR_WIDTH_MOBILE`, `SIDEBAR_WIDTH_ICON`.

Primary location	<code>apps/web/client/src/components/ui/sidebar.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — skeleton.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/skeleton.tsx` and its visible exported or local symbols: `Skeleton`.

Primary location	<code>apps/web/client/src/components/ui/skeleton.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — slider.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/slider.tsx` and its visible exported or local symbols: `Slider`, `_values`.

Primary location	<code>apps/web/client/src/components/ui/slider.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — sonner.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/sonner.tsx` and its visible exported or local symbols: `Toaster`.

Primary location	<code>apps/web/client/src/components/ui/sonner.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — spinner.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/spinner.tsx` and its visible exported or local symbols: `Spinner`.

Primary location	<code>apps/web/client/src/components/ui/spinner.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — switch.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/switch.tsx` and its visible exported or local symbols: `Switch`.

Primary location	<code>apps/web/client/src/components/ui/switch.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — table.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/table.tsx` and its visible exported or local symbols: `Table`, `TableHeader`, `TableBody`, `TableFooter`, `TableRow`.

Primary location	<code>apps/web/client/src/components/ui/table.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — tabs.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/tabs.tsx` and its visible exported or local symbols: `Tabs`, `TabsList`, `TabsTrigger`, `TabsContent`.

Primary location	<code>apps/web/client/src/components/ui/tabs.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — textarea.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/textarea.tsx` and its visible exported or local symbols: `Textarea`, `dialogComposition`, `isComposing`.

Primary location	<code>apps/web/client/src/components/ui/textarea.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — toggle-group.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/toggle-group.tsx` and its visible exported or local symbols: `ToggleGroupContext`, `ToggleGroup`, `ToggleGroupItem`, `context`.

Primary location	<code>apps/web/client/src/components/ui/toggle-group.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — toggle.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/toggle.tsx` and its visible exported or local symbols: `toggleVariants`, `Toggle`.

Primary location	<code>apps/web/client/src/components/ui/toggle.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — tooltip.tsx

shared interface component

Purpose. This sheet documents the real repository unit `apps/web/client/src/components/ui/tooltip.tsx` and its visible exported or local symbols: `TooltipProvider`, `Tooltip`, `TooltipTrigger`, `TooltipContent`.

Primary location	<code>apps/web/client/src/components/ui/tooltip.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Preserve its public props and test every calling page after a change.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — const.ts

client implementation module

Purpose. This sheet documents the real repository unit `apps/web/client/src/const.ts` and its visible exported or local symbols: `startLogin`, `oauthPortalUrl`, `appId`, `redirectUri`, `nonce`.

Primary location	<code>apps/web/client/src/const.ts</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Use a narrow change and trace all imports before refactoring.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — ThemeContext.tsx

client implementation module

Purpose. This sheet documents the real repository unit `apps/web/client/src/contexts/ThemeContext.tsx` and its visible exported or local symbols: `ThemeContext`, `ThemeProvider`, `stored`, `root`, `toggleTheme`.

Primary location	<code>apps/web/client/src/contexts/ThemeContext.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Use a narrow change and trace all imports before refactoring.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — marketplace.ts

client implementation module

Purpose. This sheet documents the real repository unit `apps/web/client/src/data/marketplace.ts` and its visible exported or local symbols: `requestPreview`, `flightPreview`, `categories`.

Primary location	<code>apps/web/client/src/data/marketplace.ts</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Use a narrow change and trace all imports before refactoring.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — useComposition.ts

shared client utility

Purpose. This sheet documents the real repository unit `apps/web/client/src/hooks/useComposition.ts` and its visible exported or local symbols: `useComposition`, `c`, `timer`, `timer2`, `onCompositionStart`.

Primary location	<code>apps/web/client/src/hooks/useComposition.ts</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Avoid creating unbounded browser work or storing private data in browser persistence.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — useMobile.tsx

shared client utility

Purpose. This sheet documents the real repository unit `apps/web/client/src/hooks/useMobile.tsx` and its visible exported or local symbols: `MOBILE_BREAKPOINT`, `useIsMobile`, `mql`, `onChange`.

Primary location	<code>apps/web/client/src/hooks/useMobile.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Avoid creating unbounded browser work or storing private data in browser persistence.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — usePersistFn.ts

shared client utility

Purpose. This sheet documents the real repository unit `apps/web/client/src/hooks/usePersistFn.ts` and its visible exported or local symbols: `usePersistFn`, `fnRef`, `persistFn`.

Primary location	<code>apps/web/client/src/hooks/usePersistFn.ts</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Avoid creating unbounded browser work or storing private data in browser persistence.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — currencies.ts

shared client utility

Purpose. This sheet documents the real repository unit `apps/web/client/src/lib/currencies.ts` and its visible exported or local symbols: `SUPPORTED_CURRENCIES`, `currencyLabel`.

Primary location	<code>apps/web/client/src/lib/currencies.ts</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Avoid creating unbounded browser work or storing private data in browser persistence.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — feedback.ts

shared client utility

Purpose. This sheet documents the real repository unit `apps/web/client/src/lib/feedback.ts` and its visible exported or local symbols: `feedbackPatterns`, `feedbackSymbols`, `showBridgeXVisualFeedback`, `cue`, `playBridgeXFeedback`.

Primary location	<code>apps/web/client/src/lib/feedback.ts</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Avoid creating unbounded browser work or storing private data in browser persistence.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — fileUpload.ts

shared client utility

Purpose. This sheet documents the real repository unit `apps/web/client/src/lib/fileUpload.ts` and its visible exported or local symbols: `MAX_DIMENSION`, `MAX_VIDEO_SECONDS`, `MAX_VIDEO_BYTES`, `compressImageForUpload`, `source`.

Primary location	<code>apps/web/client/src/lib/fileUpload.ts</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Avoid creating unbounded browser work or storing private data in browser persistence.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — supabase.ts

shared client utility

Purpose. This sheet documents the real repository unit `apps/web/client/src/lib/supabase.ts` and its visible exported or local symbols: `supabase`.

Primary location	<code>apps/web/client/src/lib/supabase.ts</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Avoid creating unbounded browser work or storing private data in browser persistence.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — trpc.ts

shared client utility

Purpose. This sheet documents the real repository unit `apps/web/client/src/lib/trpc.ts` and its visible exported or local symbols: `trpc`.

Primary location	<code>apps/web/client/src/lib/trpc.ts</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Avoid creating unbounded browser work or storing private data in browser persistence.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — utils.ts

shared client utility

Purpose. This sheet documents the real repository unit `apps/web/client/src/lib/utils.ts` and its visible exported or local symbols: `cn`.

Primary location	<code>apps/web/client/src/lib/utils.ts</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Avoid creating unbounded browser work or storing private data in browser persistence.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — main.tsx

client implementation module

Purpose. This sheet documents the real repository unit `apps/web/client/src/main.tsx` and its visible exported or local symbols: `queryClient`, `recoveryKey`, `redirectToLoginIfUnauthorized`, `isUnauthorized`, `error`.

Primary location	<code>apps/web/client/src/main.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Use a narrow change and trace all imports before refactoring.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — Access.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/Access.tsx` and its visible exported or local symbols: `Access`, `reportError`, `submit`, `signInWithGoogle`.

Primary location	<code>apps/web/client/src/pages/Access.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — AdminChats.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/AdminChats.tsx` and its visible exported or local symbols: `time`, `AdminChats`, `selectedId`, `load`, `rows`.

Primary location	<code>apps/web/client/src/pages/AdminChats.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — AdminControl.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/AdminControl.tsx` and its visible exported or local symbols: `tabs`, `PAGE_SIZE`, `tabForPath`, `segment`, `tableByTab`.

Primary location	<code>apps/web/client/src/pages/AdminControl.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — AdminEnquiries.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/AdminEnquiries.tsx` and its visible exported or local symbols: `messageTime`, `AdminEnquiries`, `timer`, `loadMessages`, `load`.

Primary location	<code>apps/web/client/src/pages/AdminEnquiries.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — AdminExchangeRates.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/AdminExchangeRates.tsx` and its visible exported or local symbols: `currencies`, `AdminExchangeRates`, `load`, `selected`, `save`.

Primary location	<code>apps/web/client/src/pages/AdminExchangeRates.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — AdminPaymentReview.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/AdminPaymentReview.tsx` and its visible exported or local symbols: `money`, `statusLabel`, `memberLabel`, `AdminPaymentReview`, `paymentId`.

Primary location	<code>apps/web/client/src/pages/AdminPaymentReview.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — AdminVerificationPerson.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/AdminVerificationPerson.tsx` and its visible exported or local symbols: `AdminVerificationPerson`, `userId`, `load`, `verified`, `reviewAll`.

Primary location	<code>apps/web/client/src/pages/AdminVerificationPerson.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — ComponentShowcase.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/ComponentShowcase.tsx` and its visible exported or local symbols: `ComponentsShowcase`, `handleDialogSubmit`, `handleDialogKeyDown`, `handleChatSend`, `newMessages`.

Primary location	<code>apps/web/client/src/pages/ComponentShowcase.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — ContactPage.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/ContactPage.tsx` and its visible exported or local symbols: `ContactPage`, `submit`.

Primary location	<code>apps/web/client/src/pages/ContactPage.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — CreateFlow.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/CreateFlow.tsx` and its visible exported or local symbols: `SERVICE_TYPES`, `PRODUCT_CATEGORIES`, `QUANTITY_AWARE_ITEMS`, `friendlyError`, `text`.

Primary location	<code>apps/web/client/src/pages/CreateFlow.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — Deals.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/Deals.tsx` and its visible exported or local symbols: `statusLabel`, `displayTime`, `date`, `isSupportUpdate`, `playSendTone`.

Primary location	<code>apps/web/client/src/pages/Deals.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — GlobalVerification.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/GlobalVerification.tsx` and its visible exported or local symbols: `requiredDocuments`, `GlobalVerification`, `attach`, `setStudentStatus`, `submit`.

Primary location	<code>apps/web/client/src/pages/GlobalVerification.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — Home.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/Home.tsx` and its visible exported or local symbols: `Home`, `allRows`, `ownerIds`, `badgeById`, `ratingById`.

Primary location	<code>apps/web/client/src/pages/Home.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — InterestPage.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/InterestPage.tsx` and its visible exported or local symbols: `CATEGORIES`, `QUANTITY_AWARE_ITEMS`, `InterestPage`, `nextListing`, `currency`.

Primary location	<code>apps/web/client/src/pages/InterestPage.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — ManagePosts.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/ManagePosts.tsx` and its visible exported or local symbols: `ManagePosts`, `requestedEditId`, `load`, `requestPosts`, `listingPosts`.

Primary location	<code>apps/web/client/src/pages/ManagePosts.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — MarketingPages.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/MarketingPages.tsx` and its visible exported or local symbols: `flow`, `faqs`, `Intro`, `HowItWorks`, `SafetyPage`.

Primary location	<code>apps/web/client/src/pages/MarketingPages.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — Marketplace.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/Marketplace.tsx` and its visible exported or local symbols: `PAGE_SIZE`, `pathsFor`, `Marketplace`, `marketplaceParams`, `initialView`.

Primary location	<code>apps/web/client/src/pages/Marketplace.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — MemberProfile.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/MemberProfile.tsx` and its visible exported or local symbols: `MemberProfile`, `memberId`, `average`, `count`.

Primary location	<code>apps/web/client/src/pages/MemberProfile.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — NotFound.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/NotFound.tsx` and its visible exported or local symbols: `NotFound`, `handleGoHome`.

Primary location	<code>apps/web/client/src/pages/NotFound.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — NotificationsPage.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/NotificationsPage.tsx` and its visible exported or local symbols: `PAGE_SIZE`, `NotificationsPage`, `load`, `page`, `markAllRead`.

Primary location	<code>apps/web/client/src/pages/NotificationsPage.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — OfferPage.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/OfferPage.tsx` and its visible exported or local symbols: `OfferPage`, `requestId`, `currency`, `submit`, `profile`.

Primary location	<code>apps/web/client/src/pages/OfferPage.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — Onboarding.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/Onboarding.tsx` and its visible exported or local symbols: `Onboarding`, `submit`.

Primary location	<code>apps/web/client/src/pages/Onboarding.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — PaymentHistory.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/PaymentHistory.tsx` and its visible exported or local symbols: `money`, `label`, `tone`, `nativeInput`, `SettlementCard`.

Primary location	<code>apps/web/client/src/pages/PaymentHistory.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — PostDetail.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/PostDetail.tsx` and its visible exported or local symbols: `money`, `mediaPaths`, `PostDetail`, `kind`, `postId`.

Primary location	<code>apps/web/client/src/pages/PostDetail.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — ReportIncident.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/ReportIncident.tsx` and its visible exported or local symbols: `ReportIncident`, `selectEvidence`, `submit`, `evidencePaths`, `file`.

Primary location	<code>apps/web/client/src/pages/ReportIncident.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — ResetPassword.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/ResetPassword.tsx` and its visible exported or local symbols: `ResetPassword`, `save`.

Primary location	<code>apps/web/client/src/pages/ResetPassword.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — ReviewsPage.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/ReviewsPage.tsx` and its visible exported or local symbols: `ReviewsPage`, `load`, `rows`, `otherIds`, `submit`.

Primary location	<code>apps/web/client/src/pages/ReviewsPage.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — SuperAdminControl.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/SuperAdminControl.tsx` and its visible exported or local symbols: `SuperAdminControl`, `load`, `visible`, `setRole`, `setSuspended`.

Primary location	<code>apps/web/client/src/pages/SuperAdminControl.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

Source map — Workspace.tsx

member-facing or administrator page

Purpose. This sheet documents the real repository unit `apps/web/client/src/pages/Workspace.tsx` and its visible exported or local symbols: `emptyMemberData`, `userName`, `formatMoney`, `statusLabel`, `CURRENCIES`.

Primary location	<code>apps/web/client/src/pages/Workspace.tsx</code>
Inputs / prerequisites	Caller route, provider state, authenticated user when required, and real database response
Expected outputs	Accessible state, UI transition, or shared helper result appropriate to the route
Security or reliability boundary	Confirm guest, member, administrator, and super-admin states before changing controls.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Run the focused regression coverage, TypeScript validation, and the relevant member/admin flow after modifying this file.

Scaling note. At scale, profile bundle size, render frequency, response cardinality, and error boundaries for this unit. Keep public lists paginated and do not move private record filtering into the browser.

3 Volume III — Database, Storage, and Migration Ledger

BridgeX production schema changes are forward-only migrations. Each entry below is derived from a real migration filename and should be read alongside the live database definition before a follow-up migration is written.

Migration ledger — Post Media Galleries

Forward-only database change

Purpose. Migration 202608181305_post_media_galleries.sql introduces or adjusts request_media_read_when_open_post.

Primary location	supabase/migrations/202608181305_post_media_galleries.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Post Media Galleries

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181305_post_media_galleries.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Notifications And Push Tokens

Forward-only database change

Purpose. Migration 202608181315_notifications_and_push_tokens.sql introduces or adjusts public, public, notifications_read_own_or_admin, notifications_insert_related_or_admin, notifications_update_own_or_admin.

Primary location	supabase/migrations/202608181315_notifications_and_push_tokens.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Notifications And Push Tokens

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181315_notifications_and_push_tokens.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Member Suspension Enforcement

Forward-only database change

Purpose. Migration `202608181325\_member\_suspension\_enforcement.sql` introduces or adjusts `send_requests_insert_own`, `carry_listings_insert_own`, `offers_insert_own`, `listing_interests_insert_own`, `verification_insert_own`.

Primary location	<code>supabase/migrations/202608181325_member_suspension_enforcement.sql</code>
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Member Suspension Enforcement

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181325_member_suspension_enforcement.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Protected Matches Chat Guest Visibility

Forward-only database change

Purpose. Migration 202608181425_protected_matches_chat_guest_visibility.sql introduces or adjusts public, public, send_requests_read_open_or_owner, carry_listings_read_open_or_owner, matches_read_participant_or_admin.

Primary location	supabase/ migrations/202608181425_protected_matches_chat_guest_visibility.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Protected Matches Chat Guest Visibility

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/ migrations/202608181425_protected_matches_chat_guest_visibility.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Expo Push Delivery

Forward-only database change

Purpose. Migration 202608181435_expo_push_delivery.sql introduces or adjusts bridgeX_dispatch_expo_push.

Primary location	supabase/migrations/202608181435_expo_push_delivery.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Expo Push Delivery

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181435_expo_push_delivery.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Super Admin Governance

Forward-only database change

Purpose. Migration 202608181545_super_admin_governance.sql introduces or adjusts bridgeX_protect_administrator_hierarchy.

Primary location	supabase/migrations/202608181545_super_admin_governance.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Super Admin Governance

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181545_super_admin_governance.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Order Progress And Contacts

Forward-only database change

Purpose. Migration 202608181600_order_progress_and_contacts.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608181600_order_progress_and_contacts.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Order Progress And Contacts

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181600_order_progress_and_contacts.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Grant Reminder Scheduler

Forward-only database change

Purpose. Migration 202608181610_grant_reminder_scheduler.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608181610_grant_reminder_scheduler.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Grant Reminder Scheduler

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181610_grant_reminder_scheduler.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Fix Order Fulfillment States

Forward-only database change

Purpose. Migration `202608181635\_fix\_order\_fulfillment\_states.sql` introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608181635_fix_order_fulfillment_states.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Fix Order Fulfillment States

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181635_fix_order_fulfillment_states.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Multi Category Requests

Forward-only database change

Purpose. Migration 202608181700_multi_category_requests.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608181700_multi_category_requests.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Multi Category Requests

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181700_multi_category_requests.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Match Inbox Metadata

Forward-only database change

Purpose. Migration `202608181715\_match\_inbox\_metadata.sql` introduces or adjusts `bridgex_snapshot_match_names_before_insert`.

Primary location	supabase/migrations/202608181715_match_inbox_metadata.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Match Inbox Metadata

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181715_match_inbox_metadata.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Match Last Message Snapshot

Forward-only database change

Purpose. Migration `202608181730\_match\_last\_message\_snapshot.sql` introduces or adjusts `bridgex_snapshot_match_last_message_after_insert`.

Primary location	<code>supabase/migrations/202608181730_match_last_message_snapshot.sql</code>
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Match Last Message Snapshot

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181730_match_last_message_snapshot.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Match Verified Snapshots

Forward-only database change

Purpose. Migration 202608181740_match_verified_snapshots.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608181740_match_verified_snapshots.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Match Verified Snapshots

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181740_match_verified_snapshots.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Public Member Badges

Forward-only database change

Purpose. Migration 202608181900_public_member_badges.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608181900_public_member_badges.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Public Member Badges

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181900_public_member_badges.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Admin Order Actions

Forward-only database change

Purpose. Migration 202608181910_admin_order_actions.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608181910_admin_order_actions.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Admin Order Actions

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181910_admin_order_actions.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Password Reset Assistance

Forward-only database change

Purpose. Migration 202608181930_password_reset_assistance.sql introduces or adjusts public, password_reset_logs_super_admin_read.

Primary location	supabase/migrations/202608181930_password_reset_assistance.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Password Reset Assistance

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608181930_password_reset_assistance.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Sender Delivery Release And Media Cleanup

Forward-only database change

Purpose. Migration 202608181945_sender_delivery_release_and_media_cleanup.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/ migrations/202608181945_sender_delivery_release_and_media_cleanup.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Sender Delivery Release And Media Cleanup

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/ migrations/202608181945_sender_delivery_release_and_media_cleanup.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Request Route Deadline And Handling

Forward-only database change

Purpose. Migration `202608182000\_request\_route\_deadline\_and\_handling.sql` introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/ migrations/202608182000_request_route_deadline_and_handling.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Request Route Deadline And Handling

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/ migrations/202608182000_request_route_deadline_and_handling.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Contact Enquiries

Forward-only database change

Purpose. Migration `202608182015\_contact\_enquiries.sql` introduces or adjusts `public`, `contact_enquiries_public_submit`, `contact_enquiries_admin_read`, `contact_enquiries_admin_update`.

Primary location	<code>supabase/migrations/202608182015_contact_enquiries.sql</code>
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Contact Enquiries

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608182015_contact_enquiries.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Contact Enquiry Replies

Forward-only database change

Purpose. Migration `202608182030\_contact\_enquiry\_replies.sql` introduces or adjusts `contact_enquiries_member_read_own`.

Primary location	supabase/migrations/202608182030_contact_enquiry_replies.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Contact Enquiry Replies

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608182030_contact_enquiry_replies.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Completed Order Reviews

Forward-only database change

Purpose. Migration 202608182045_completed_order_reviews.sql introduces or adjusts public, completed_reviews_public_read, completed_reviews_participant_insert.

Primary location	supabase/migrations/202608182045_completed_order_reviews.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Completed Order Reviews

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608182045_completed_order_reviews.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Carry Capacity And Interest Details

Forward-only database change

Purpose. Migration `202608182100\_carry\_capacity\_and\_interest\_details.sql` introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/ migrations/202608182100_carry_capacity_and_interest_details.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Carry Capacity And Interest Details

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/ migrations/202608182100_carry_capacity_and_interest_details.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Cargo Capacity Fulfillment

Forward-only database change

Purpose. Migration 202608190900_cargo_capacity_fulfillment.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608190900_cargo_capacity_fulfillment.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Cargo Capacity Fulfillment

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608190900_cargo_capacity_fulfillment.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Interest Item Quantities And Delivery Date

Forward-only database change

Purpose. Migration 202608190930_interest_item_quantities_and_delivery_date.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/ migrations/202608190930_interest_item_quantities_and_delivery_date.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Interest Item Quantities And Delivery Date

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/ migrations/202608190930_interest_item_quantities_and_delivery_date.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Secure Contact Reply Delivery

Forward-only database change

Purpose. Migration `202608191000\_secure\_contact\_reply\_delivery.sql` introduces or adjusts schema, policy, function, or trigger changes.

Primary location	<code>supabase/migrations/202608191000_secure_contact_reply_delivery.sql</code>
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Secure Contact Reply Delivery

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191000_secure_contact_reply_delivery.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Admin Section Read State

Forward-only database change

Purpose. Migration 202608191015_admin_section_read_state.sql introduces or adjusts public, bridgex_admin_section_reads_select_own, bridgex_admin_section_reads_insert_own, bridgex_admin_section_reads_update_own.

Primary location	supabase/migrations/202608191015_admin_section_read_state.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Admin Section Read State

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191015_admin_section_read_state.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Support Conversations

Forward-only database change

Purpose. Migration 202608191100_support_conversations.sql introduces or adjusts public, contact_enquiry_messages_select_participants.

Primary location	supabase/migrations/202608191100_support_conversations.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Support Conversations

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191100_support_conversations.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Fix Match Read State

Forward-only database change

Purpose. Migration `202608191115\_fix\_match\_read\_state.sql` introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608191115_fix_match_read_state.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Fix Match Read State

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191115_fix_match_read_state.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Currency Preferences And Partial Cargo Acceptance

Forward-only database change

Purpose. Migration 202608191200_currency_preferences_and_partial_cargo_acceptance.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608191200_currency_preferences_and_partial_cargo_acceptance.s
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Currency Preferences And Partial Cargo Acceptance

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191200_currency_preferences_and_partial_cargo_acceptance.s
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Carry Interest Delivery Contacts

Forward-only database change

Purpose. Migration 202608191330_carry_interest_delivery_contacts.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608191330_carry_interest_delivery_contacts.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Carry Interest Delivery Contacts

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191330_carry_interest_delivery_contacts.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Secure Protected Rpc Execution

Forward-only database change

Purpose. Migration 202608191345_secure_protected_rpc_execution.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608191345_secure_protected_rpc_execution.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Secure Protected Rpc Execution

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191345_secure_protected_rpc_execution.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Contact Enquiries Realtime

Forward-only database change

Purpose. Migration 202608191500_contact_enquiries_realtime.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608191500_contact_enquiries_realtime.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Contact Enquiries Realtime

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191500_contact_enquiries_realtime.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Match Messages Realtime

Forward-only database change

Purpose. Migration 202608191520_match_messages_realtime.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608191520_match_messages_realtime.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Match Messages Realtime

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191520_match_messages_realtime.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Manual Payment Verification

Forward-only database change

Purpose. Migration `202608191600\_manual\_payment\_verification.sql` introduces or adjusts `public`, `bridgex_manual_qr_cny_only`, `bridgex_payment_proofs_read_participant_or_admin`, `bridgex_payment_proofs_admin_update`, `payment_proofs_upload_own`.

Primary location	<code>supabase/migrations/202608191600_manual_payment_verification.sql</code>
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Manual Payment Verification

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191600_manual_payment_verification.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Cny Manual Qr Safety

Forward-only database change

Purpose. Migration 202608191615_cny_manual_qr_safety.sql introduces or adjusts bridgex_manual_qr_cny_only.

Primary location	supabase/migrations/202608191615_cny_manual_qr_safety.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Cny Manual Qr Safety

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191615_cny_manual_qr_safety.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Private Payment Instruction Assets

Forward-only database change

Purpose. Migration 202608191620_private_payment_instruction_assets.sql introduces or adjusts payment_instructions_authenticated_read.

Primary location	supabase/ migrations/202608191620_private_payment_instruction_assets.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Private Payment Instruction Assets

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/ migrations/202608191620_private_payment_instruction_assets.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Private Payment Instruction Read Scope

Forward-only database change

Purpose. Migration 202608191625_private_payment_instruction_read_scope.sql introduces or adjusts payment_instructions_payment_payer_read.

Primary location	supabase/ migrations/202608191625_private_payment_instruction_read_scope.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Private Payment Instruction Read Scope

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/ migrations/202608191625_private_payment_instruction_read_scope.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Payment Response Statuses

Forward-only database change

Purpose. Migration 202608191640_payment_response_statuses.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608191640_payment_response_statuses.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Payment Response Statuses

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191640_payment_response_statuses.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Send Request Payment Pending

Forward-only database change

Purpose. Migration `202608191700\_send\_request\_payment\_pending.sql` introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608191700_send_request_payment_pending.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Send Request Payment Pending

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191700_send_request_payment_pending.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Allow Manual Qr All Currencies

Forward-only database change

Purpose. Migration `202608191730\_allow\_manual\_qr\_all\_currencies.sql` introduces or adjusts schema, policy, function, or trigger changes.

Primary location	<code>supabase/migrations/202608191730_allow_manual_qr_all_currencies.sql</code>
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Allow Manual Qr All Currencies

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191730_allow_manual_qr_all_currencies.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Traveler Payout History

Forward-only database change

Purpose. Migration `202608191800\_traveler\_payout\_history.sql` introduces or adjusts `public, public, traveler_payout_profile_owner_or_admin_read, traveler_payout_profile_owner_write, traveler_payout_profile_owner_update`.

Primary location	supabase/migrations/202608191800_traveler_payout_history.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Traveler Payout History

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191800_traveler_payout_history.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Carry Payment Pending And Single Payment Alert

Forward-only database change

Purpose. Migration 202608191900_carry_payment_pending_and_single_payment_alert.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608191900_carry_payment_pending_and_single_payment_alert.s
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Carry Payment Pending And Single Payment Alert

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608191900_carry_payment_pending_and_single_payment_alert.s
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Live Carry Inventory And Payout Navigation

Forward-only database change

Purpose. Migration 202608192000_live_carry_inventory_and_payout_navigation.sql introduces or adjusts bridgex_confirmed_interest_inventory, bridgex_carry_listing_visibility.

Primary location	supabase/migrations/202608192000_live_carry_inventory_and_payout_navigation.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Live Carry Inventory And Payout Navigation

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608192000_live_carry_inventory_and_payout_navigation.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Completed Order Review Completion Policy

Forward-only database change

Purpose. Migration 202608200910_completed_order_review_completion_policy.sql introduces or adjusts completed_reviews_participant_insert.

Primary location	supabase/ migrations/202608200910_completed_order_review_completion_policy.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Completed Order Review Completion Policy

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/ migrations/202608200910_completed_order_review_completion_policy.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Admin Exchange Rates And Payment Snapshots

Forward-only database change

Purpose. Migration 202608201500_admin_exchange_rates_and_payment_snapshots.sql introduces or adjusts public, bridgex_exchange_rates_authenticated_read, bridgex_payment_conversion_snapshot.

Primary location	supabase/migrations/202608201500_admin_exchange_rates_and_payment_snapshots.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Admin Exchange Rates And Payment Snapshots

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608201500_admin_exchange_rates_and_payment_snapshots.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Expand Currencies And Payout Notification Recipients

Forward-only database change

Purpose. Migration 202608201700_expand_currencies_and_payout_notification_recipients.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608201700_expand_currencies_and_payout_notification_recipients.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Expand Currencies And Payout Notification Recipients

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608201700_expand_currencies_and_payout_notification_recipient
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Single Active Traveler Offer

Forward-only database change

Purpose. Migration 202608201800_single_active_traveler_offer.sql introduces or adjusts schema, policy, function, or trigger changes.

Primary location	supabase/migrations/202608201800_single_active_traveler_offer.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Single Active Traveler Offer

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608201800_single_active_traveler_offer.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

Migration ledger — Harden Payment Proof Storage And Submission

Forward-only database change

Purpose. Migration 202608201900_harden_payment_proof_storage_and_submission.sql introduces or adjusts payment_proofs_admin_read.

Primary location	supabase/migrations/202608201900_harden_payment_proof_storage_and_submission.sql
Inputs / prerequisites	A reviewed migration, dependency order, compatible client states, and an explicit rollback/forward-fix plan
Expected outputs	A reproducible production schema transition and matching application behavior
Security or reliability boundary	Never edit an applied production migration. RLS and private-storage changes require negative authorization testing.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Apply only after a staging test; verify the exact live definition, policy behavior, and affected RPC/trigger results.

Scaling note. At scale, review lock duration, index creation behavior, trigger cost, and backfill strategy. Prefer additive columns, batched backfills, and gradual cutovers.

Migration runbook — Harden Payment Proof Storage And Submission

Diagnosis and safe follow-up

Purpose. This runbook describes how to investigate a failure involving the change represented by {migration.name}.

Primary location	supabase/migrations/202608201900_harden_payment_proof_storage_and_submission.sql
Inputs / prerequisites	Incident timestamp, affected record ID, user role, exact SQL/RPC error, and before/after record state
Expected outputs	A minimal corrective migration or verified application correction
Security or reliability boundary	Do not repair a data-integrity incident by weakening an RLS policy, disabling a constraint, or manually exposing private storage.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the expected state transition, notification recipient, storage authorization, and regression test coverage.

Scaling note. At scale, capture query plan, database load, and the count of affected rows before choosing a remediation approach.

4 Volume IV — Protected Workflow, Payment, Payout, and Messaging Operations

These sheets define the operational controls for request offers, carry interests, payment proof, protected contact release, delivery confirmation, payout tracking, support, notifications, reviews, and reporting.

Guest Marketplace Discovery — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the guest marketplace discovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Guest Marketplace Discovery – Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the guest marketplace discovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Guest Marketplace Discovery — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the guest marketplace discovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Guest Marketplace Discovery – Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the guest marketplace discovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Guest Marketplace Discovery – Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the guest marketplace discovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Guest Marketplace Discovery – Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the guest marketplace discovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Guest Marketplace Discovery – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the guest marketplace discovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Guest Marketplace Discovery — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the guest marketplace discovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Guest Marketplace Discovery – Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the guest marketplace discovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Guest Marketplace Discovery — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the guest marketplace discovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Onboarding — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the member onboarding domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Onboarding – Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the member onboarding domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Onboarding — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the member onboarding domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Onboarding — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the member onboarding domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Onboarding — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the member onboarding domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Onboarding — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the member onboarding domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Onboarding — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the member onboarding domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Onboarding — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the member onboarding domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Onboarding — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the member onboarding domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Onboarding — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the member onboarding domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Identity Verification — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the identity verification domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Identity Verification — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the identity verification domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Identity Verification — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the identity verification domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Identity Verification — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the identity verification domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Identity Verification — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the identity verification domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Identity Verification — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the identity verification domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Identity Verification — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the identity verification domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Identity Verification — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the identity verification domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Identity Verification — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the identity verification domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Identity Verification — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the identity verification domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Send-Request Publication — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the send-request publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Send-Request Publication — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the `send-request` publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Send-Request Publication — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the send-request publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Send-Request Publication — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the send-request publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Send-Request Publication — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the `send-request` publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Send-Request Publication — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the send-request publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Send-Request Publication – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the send-request publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Send-Request Publication — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the send-request publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Send-Request Publication — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the send-request publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Send-Request Publication — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the send-request publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Carry-Listing Publication — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the carry-listing publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Carry-Listing Publication — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the carry-listing publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Carry-Listing Publication — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the carry-listing publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Carry-Listing Publication — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the carry-listing publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Carry-Listing Publication — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the carry-listing publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Carry-Listing Publication — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the carry-listing publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Carry-Listing Publication – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the carry-listing publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Carry-Listing Publication — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the carry-listing publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Carry-Listing Publication — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the carry-listing publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Carry-Listing Publication — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the carry-listing publication domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Offer — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the `traveler offer` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Offer — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the `traveler offer` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Offer — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the `traveler offer` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Offer — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the `traveler offer` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Offer — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the `traveler offer` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Offer — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the `traveler offer` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Offer — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the traveler offer domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Offer — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the traveler offer domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Offer — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the `traveler offer` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Offer — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the `traveler offer` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Sender Interest — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the sender interest domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Sender Interest — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the sender interest domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Sender Interest — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the sender interest domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Sender Interest — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the sender interest domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Sender Interest — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the `sender interest` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Sender Interest — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the sender interest domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Sender Interest — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the sender interest domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Sender Interest — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the sender interest domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Sender Interest — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the sender interest domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Sender Interest — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the `sender_interest` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Single Active Traveler Selection — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the `single active traveler selection` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Single Active Traveler Selection — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the `single active traveler selection` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Single Active Traveler Selection — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the single active traveler selection domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Single Active Traveler Selection — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the `single active traveler selection` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Single Active Traveler Selection — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the single active traveler selection domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Single Active Traveler Selection — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the single active traveler selection domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Single Active Traveler Selection — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the `single active traveler selection` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Single Active Traveler Selection — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the single active traveler selection domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Single Active Traveler Selection — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the `single active traveler selection` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Single Active Traveler Selection — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the single active traveler selection domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Request — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the payment request domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Request — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the payment request domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Request — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the payment request domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Request — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the payment request domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Request — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the payment request domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Request — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the payment request domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Request — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the payment request domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Request — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the payment request domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Request — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the payment request domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Request — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the payment request domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Proof Upload — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the payment proof upload domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Proof Upload — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the payment proof upload domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Proof Upload — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the payment proof upload domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Proof Upload — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the payment proof upload domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Proof Upload — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the payment proof upload domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Proof Upload — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the payment proof upload domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Proof Upload – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the payment proof upload domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Proof Upload — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the payment proof upload domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Proof Upload – Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the payment proof upload domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payment Proof Upload — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the payment proof upload domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Payment Review — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the administrator payment review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Payment Review — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the administrator payment review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Payment Review — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the `administrator payment review` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Payment Review — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the `administrator payment review` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Payment Review — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the `administrator payment review` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Payment Review — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the administrator payment review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Payment Review — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the administrator payment review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Payment Review — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the `administrator payment review` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Payment Review — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the administrator payment review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Payment Review — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the `administrator payment review` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Exchange-Rate Snapshot — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the exchange-rate snapshot domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Exchange-Rate Snapshot — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the exchange-rate snapshot domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Exchange-Rate Snapshot — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the exchange-rate snapshot domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Exchange-Rate Snapshot — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the exchange-rate snapshot domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Exchange-Rate Snapshot — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the exchange-rate snapshot domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Exchange-Rate Snapshot — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the exchange-rate snapshot domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Exchange-Rate Snapshot — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the exchange-rate snapshot domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Exchange-Rate Snapshot — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the exchange-rate snapshot domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Exchange-Rate Snapshot — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the exchange-rate snapshot domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Exchange-Rate Snapshot — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the exchange-rate snapshot domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Match Opening — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the protected match opening domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Match Opening — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the protected match opening domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Match Opening — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the protected match opening domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Match Opening — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the protected match opening domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Match Opening — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the protected match opening domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Match Opening — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the protected match opening domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Match Opening – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the protected match opening domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Match Opening — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the protected match opening domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Match Opening – Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the protected match opening domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Match Opening — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the protected match opening domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Private Contact Release — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the `private contact release` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Private Contact Release — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the `private contact release` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Private Contact Release — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the private contact release domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Private Contact Release — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the `private contact release` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Private Contact Release — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the `private contact release` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Private Contact Release — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the private contact release domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Private Contact Release — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the `private contact release` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Private Contact Release — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the `private contact release` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Private Contact Release — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the `private contact release` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Private Contact Release — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the private contact release domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Deal Chat — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the protected deal chat domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Deal Chat — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the protected deal chat domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Deal Chat — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the protected deal chat domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Deal Chat — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the protected deal chat domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Deal Chat — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the protected deal chat domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Deal Chat — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the protected deal chat domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Deal Chat — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the protected deal chat domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Deal Chat — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the protected deal chat domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Deal Chat — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the protected deal chat domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Protected Deal Chat — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the protected deal chat domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Support Conversation — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the support conversation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Support Conversation — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the support conversation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Support Conversation — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the support conversation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Support Conversation — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the support conversation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Support Conversation — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the support conversation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Support Conversation — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the support conversation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Support Conversation — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the support conversation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Support Conversation — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the support conversation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Support Conversation — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the support conversation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Support Conversation — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the support conversation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Contact Enquiry — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the contact enquiry domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Contact Enquiry – Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the contact enquiry domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Contact Enquiry — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the contact enquiry domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Contact Enquiry – Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the contact enquiry domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Contact Enquiry — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the contact enquiry domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Contact Enquiry — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the contact enquiry domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Contact Enquiry – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the contact enquiry domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Contact Enquiry – Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the contact enquiry domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Contact Enquiry — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the contact enquiry domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Contact Enquiry — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the contact enquiry domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Notification Routing — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the notification routing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Notification Routing — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the notification routing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Notification Routing — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the notification routing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Notification Routing — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the notification routing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Notification Routing — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the notification routing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Notification Routing — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the notification routing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Notification Routing – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the notification routing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Notification Routing — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the notification routing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Notification Routing — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the notification routing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Notification Routing — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the notification routing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Push-Token Registration — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the push-token registration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Push-Token Registration — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the push-token registration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Push-Token Registration — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the push-token registration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Push-Token Registration — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the push-token registration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Push-Token Registration — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the push-token registration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Push-Token Registration — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the push-token registration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Push-Token Registration – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the push-token registration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Push-Token Registration — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the push-token registration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Push-Token Registration — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the push-token registration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Push-Token Registration — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the push-token registration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Order Progress — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the order progress domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Order Progress — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the order progress domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Order Progress — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the order progress domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Order Progress — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the order progress domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Order Progress — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the order progress domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Order Progress — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the order progress domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Order Progress — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the order progress domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Order Progress — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the order progress domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Order Progress — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the order progress domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Order Progress — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the order progress domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Delivery Confirmation – Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the delivery confirmation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Delivery Confirmation — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the delivery confirmation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Delivery Confirmation — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the delivery confirmation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Delivery Confirmation — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the delivery confirmation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Delivery Confirmation — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the delivery confirmation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Delivery Confirmation — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the delivery confirmation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Delivery Confirmation — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the delivery confirmation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Delivery Confirmation — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the delivery confirmation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Delivery Confirmation — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the delivery confirmation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Delivery Confirmation — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the delivery confirmation domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Profile — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the `traveler payout profile` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Profile — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the traveler payout profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Profile — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the traveler payout profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Profile — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the `traveler payout profile` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Profile — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the `traveler payout profile` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Profile — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the traveler payout profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Profile – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the `traveler payout profile` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Profile — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the traveler payout profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Profile — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the `traveler payout profile` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Profile — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the traveler payout profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Due — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the `traveler payout due` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Due — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the `traveler payout due` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Due — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the traveler payout due domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Due — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the `traveler payout due` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Due — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the `traveler payout due` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Due — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the `traveler payout due` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Due – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the `traveler payout due` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Due — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the `traveler payout due` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Due — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the `traveler payout due` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Traveler Payout Due — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the `traveler payout due` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payout Receipt — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the payout receipt domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payout Receipt — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the payout receipt domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payout Receipt — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the payout receipt domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payout Receipt — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the payout receipt domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payout Receipt — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the payout receipt domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payout Receipt — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the payout receipt domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payout Receipt – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the payout receipt domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payout Receipt — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the payout receipt domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payout Receipt — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the payout receipt domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Payout Receipt — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the payout receipt domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Completed-Order Review — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the `completed-order review` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Completed-Order Review — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the `completed-order` review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Completed-Order Review — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the completed-order review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Completed-Order Review — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the `completed-order review` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Completed-Order Review — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the completed-order review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Completed-Order Review — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the completed-order review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Completed-Order Review — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the completed-order review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Completed-Order Review — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the `completed-order` review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Completed-Order Review — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the `completed-order` review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Completed-Order Review — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the completed-order review domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Report — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the member report domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Report – Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the member report domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Report — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the member report domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Report — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the member report domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Report — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the member report domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Report — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the member report domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Report — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the member report domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Report — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the member report domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Report — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the member report domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Member Report — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the member report domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Governance — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the administrator governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Governance — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the administrator governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Governance — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the `administrator` governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Governance — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the administrator governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Governance — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the `administrator` governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Governance — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the administrator governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Governance — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the administrator governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Governance — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the `administrator` governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Governance — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the administrator governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Administrator Governance — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the administrator governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Super-Admin Governance — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the super-admin governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Super-Admin Governance — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the super-admin governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Super-Admin Governance — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the super-admin governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Super-Admin Governance — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the super-admin governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Super-Admin Governance — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the super-admin governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Super-Admin Governance — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the super-admin governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Super-Admin Governance – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the super-admin governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Super-Admin Governance — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the super-admin governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Super-Admin Governance — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the super-admin governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Super-Admin Governance — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the super-admin governance domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Post Editing — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the post editing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Post Editing – Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the post editing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Post Editing — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the post editing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Post Editing — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the post editing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Post Editing — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the post editing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Post Editing — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the post editing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Post Editing — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the post editing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Post Editing — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the post editing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Post Editing — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the post editing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Post Editing — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the post editing domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Public Profile — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the public profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Public Profile — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the public profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Public Profile — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the public profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Public Profile — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the public profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Public Profile — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the public profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Public Profile — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the public profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Public Profile — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the public profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Public Profile — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the public profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Public Profile — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the public profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Public Profile — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the public profile domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Media Lifecycle — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the media lifecycle domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Media Lifecycle — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the media lifecycle domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Media Lifecycle — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the media lifecycle domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Media Lifecycle — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the media lifecycle domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Media Lifecycle — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the media lifecycle domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Media Lifecycle — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the media lifecycle domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Media Lifecycle — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the media lifecycle domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Media Lifecycle — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the media lifecycle domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Media Lifecycle — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the `media_lifecycle` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Media Lifecycle — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the media lifecycle domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Android Webview Shell — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the `Android WebView` shell domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Android Webview Shell — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the Android WebView shell domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Android Webview Shell — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the Android WebView shell domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Android Webview Shell — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the Android WebView shell domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Android Webview Shell — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the Android WebView shell domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Android Webview Shell — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the Android WebView shell domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Android Webview Shell – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the Android WebView shell domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Android Webview Shell — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the Android WebView shell domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Android Webview Shell — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the Android WebView shell domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Android Webview Shell — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the Android WebView shell domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Browser Recovery — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the browser recovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Browser Recovery — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the browser recovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Browser Recovery — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the browser recovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Browser Recovery — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the browser recovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Browser Recovery — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the browser recovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Browser Recovery — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the browser recovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Browser Recovery – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the browser recovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Browser Recovery — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the browser recovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Browser Recovery — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the browser recovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Browser Recovery — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the browser recovery domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Render Deployment — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the `Render deployment` domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Render Deployment — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the Render deployment domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Render Deployment — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the Render deployment domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Render Deployment — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the Render deployment domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Render Deployment — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the Render deployment domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Render Deployment — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the Render deployment domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Render Deployment – Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the Render deployment domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Render Deployment — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the Render deployment domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Render Deployment — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the Render deployment domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Render Deployment — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the Render deployment domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Supabase Storage — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the Supabase storage domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Supabase Storage — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the Supabase storage domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Supabase Storage — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the Supabase storage domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Supabase Storage — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the Supabase storage domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Supabase Storage — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the Supabase storage domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Supabase Storage — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the Supabase storage domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Supabase Storage — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the Supabase storage domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Supabase Storage — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the Supabase storage domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Supabase Storage — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the Supabase storage domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Supabase Storage — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the Supabase storage domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Database Migration — Normal Operation

Operational reference sheet

Purpose. Describe the supported member or administrator action and the expected durable state after it completes. This sheet applies the rule to the database migration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Database Migration — Data Integrity

Operational reference sheet

Purpose. Confirm every companion record, status value, audit timestamp, and related notification before declaring success. This sheet applies the rule to the database migration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Database Migration — Authorization

Operational reference sheet

Purpose. Verify both positive and negative role cases; client visibility alone is never permission. This sheet applies the rule to the database migration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Database Migration — Privacy

Operational reference sheet

Purpose. Classify each field as public, protected-match, authorized-admin, or private-storage data before it is displayed or signed. This sheet applies the rule to the database migration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Database Migration — Performance

Operational reference sheet

Purpose. Measure p95 latency, database rows scanned, payload size, and render count. Paginate and index before growing concurrency. This sheet applies the rule to the database migration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Database Migration — Failure Diagnosis

Operational reference sheet

Purpose. Start from the record ID and timestamp, then trace browser, API/RPC, database, storage, and notification outcomes in order. This sheet applies the rule to the database migration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Database Migration — Quality Assurance

Operational reference sheet

Purpose. Write a focused regression test for the business boundary and complete a role-specific manual flow on staging. This sheet applies the rule to the database migration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Database Migration — Release

Operational reference sheet

Purpose. Ship source, migration, tests, configuration notes, and a rollback/forward-fix plan as one reviewed change set. This sheet applies the rule to the database migration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Database Migration — Observability

Operational reference sheet

Purpose. Record metrics, structured error context, audit correlation, alert threshold, and an owner for the service-level objective. This sheet applies the rule to the database migration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

Database Migration — One-Million-User Scale

Operational reference sheet

Purpose. Move slow work to a queue, cache bounded public reads, protect writes with rate limits, and keep the API stateless for multiple instances. This sheet applies the rule to the database migration domain.

Primary location	Relevant page, Supabase migration, protected RPC, storage bucket, and notification record identified by the active workflow
Inputs / prerequisites	Role-scoped record, valid workflow state, and no unreviewed private-data access
Expected outputs	A bounded state transition, queue action, or documented no-op
Security or reliability boundary	Do not bypass payment review, document security, RLS, recipient routing, or rate limits to make an individual case appear to work.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A responsible engineer can reproduce the flow on staging, inspect the durable records, and demonstrate that unauthorized roles receive no private data.

Scaling note. At scale, isolate this workload from synchronous browser rendering and prevent unbounded reads, uploads, messages, or notifications from consuming shared capacity.

5 Volume V — Troubleshooting, Incident Response, QA, and Future Reliability

This volume supplies a structured problem-to-solution system. It is intentionally prescriptive about evidence, safe containment, verification, and prevention.

Troubleshooting runbook — Blank Page Or Stale Javascript Asset

Problem → cause → location → solution

Purpose. Investigate the observed symptom blank page or stale JavaScript asset with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Render 502 Routing Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom Render 502 routing failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Slow Initial Service Response

Problem → cause → location → solution

Purpose. Investigate the observed symptom `slow initial service response` with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Failed Authentication Callback

Problem → cause → location → solution

Purpose. Investigate the observed symptom failed authentication callback with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Password Reset Delivery Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom password reset delivery failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Profile Save Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom profile save failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Verification-Document Upload Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom verification-document upload failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Private Document Authorization Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom private document authorization failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Post Media Upload Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom post media upload failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Offer Acceptance Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom offer acceptance failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Interest Capacity Conflict

Problem → cause → location → solution

Purpose. Investigate the observed symptom interest capacity conflict with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Payment Record Missing

Problem → cause → location → solution

Purpose. Investigate the observed symptom payment record missing with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Payment Proof Missing In Review

Problem → cause → location → solution

Purpose. Investigate the observed symptom payment proof missing in review with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Payment Method Not Recorded

Problem → cause → location → solution

Purpose. Investigate the observed symptom payment method not recorded with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Payment Verification Rejection

Problem → cause → location → solution

Purpose. Investigate the observed symptom payment verification rejection with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Payment Qr Unavailable

Problem → cause → location → solution

Purpose. Investigate the observed symptom payment QR unavailable with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Exchange-Rate Not Published

Problem → cause → location → solution

Purpose. Investigate the observed symptom exchange-rate not published with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Protected Match Not Opened

Problem → cause → location → solution

Purpose. Investigate the observed symptom protected match not opened with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Private Contact Not Revealed

Problem → cause → location → solution

Purpose. Investigate the observed symptom private contact not revealed with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Chat Message Not Delivered

Problem → cause → location → solution

Purpose. Investigate the observed symptom chat message not delivered with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Chat Attachment Not Stored

Problem → cause → location → solution

Purpose. Investigate the observed symptom chat attachment not stored with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Notification Unread Count Mismatch

Problem → cause → location → solution

Purpose. Investigate the observed symptom notification unread count mismatch with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Push Token Registration Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom push token registration failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Workspace Data Slow

Problem → cause → location → solution

Purpose. Investigate the observed symptom workspace data slow with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Marketplace Pagination Slow

Problem → cause → location → solution

Purpose. Investigate the observed symptom marketplace pagination slow with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Public Profile Mismatch

Problem → cause → location → solution

Purpose. Investigate the observed symptom public profile mismatch with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Review Submission Rejected

Problem → cause → location → solution

Purpose. Investigate the observed symptom review submission rejected with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Rating Count Inconsistent

Problem → cause → location → solution

Purpose. Investigate the observed symptom rating count inconsistent with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Report Creation Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom report creation failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Administrator Queue Missing Record

Problem → cause → location → solution

Purpose. Investigate the observed symptom administrator queue missing record with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Super-Admin Governance Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom super-admin governance failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook – Rls Policy Denial

Problem → cause → location → solution

Purpose. Investigate the observed symptom RLS policy denial with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Database Constraint Denial

Problem → cause → location → solution

Purpose. Investigate the observed symptom database constraint denial with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Trigger Performance Regression

Problem → cause → location → solution

Purpose. Investigate the observed symptom trigger performance regression with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Migration Ordering Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom migration ordering failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Storage Signed Url Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom storage signed URL failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Mobile Gallery Selection Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom mobile gallery selection failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Android Keyboard Obstruction

Problem → cause → location → solution

Purpose. Investigate the observed symptom Android keyboard obstruction with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Android System Back Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom Android system Back failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Android State Restore Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom Android state restore failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Apk Build Authorization Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom APK build authorization failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Play App Bundle Version Conflict

Problem → cause → location → solution

Purpose. Investigate the observed symptom Play App Bundle version conflict with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Render Deployment Unavailable

Problem → cause → location → solution

Purpose. Investigate the observed symptom `Render deployment unavailable` with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Custom-Domain Certificate Issue

Problem → cause → location → solution

Purpose. Investigate the observed symptom custom-domain certificate issue with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Cdn Stale Asset Mismatch

Problem → cause → location → solution

Purpose. Investigate the observed symptom `CDN stale asset mismatch` with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Rate-Limit False Positive

Problem → cause → location → solution

Purpose. Investigate the observed symptom rate-limit false positive with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Abuse Spike

Problem → cause → location → solution

Purpose. Investigate the observed symptom abuse spike with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Malicious File Upload

Problem → cause → location → solution

Purpose. Investigate the observed symptom `malicious file upload` with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook – Queue Backlog

Problem → cause → location → solution

Purpose. Investigate the observed symptom queue backlog with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Cache Stampede

Problem → cause → location → solution

Purpose. Investigate the observed symptom cache stampede with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Database Connection Saturation

Problem → cause → location → solution

Purpose. Investigate the observed symptom database connection saturation with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Read Replica Lag

Problem → cause → location → solution

Purpose. Investigate the observed symptom `read replica lag` with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Email Delivery Delay

Problem → cause → location → solution

Purpose. Investigate the observed symptom email delivery delay with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Realtime Connection Limit

Problem → cause → location → solution

Purpose. Investigate the observed symptom realtime connection limit with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Audit Log Omission

Problem → cause → location → solution

Purpose. Investigate the observed symptom audit log omission with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Retention Deletion Failure

Problem → cause → location → solution

Purpose. Investigate the observed symptom retention deletion failure with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Backup Restoration Exercise

Problem → cause → location → solution

Purpose. Investigate the observed symptom backup restoration exercise with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Incident Communication

Problem → cause → location → solution

Purpose. Investigate the observed symptom incident communication with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook – Critical Account Restriction

Problem → cause → location → solution

Purpose. Investigate the observed symptom critical account restriction with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

Troubleshooting runbook — Legal Escalation Handoff

Problem → cause → location → solution

Purpose. Investigate the observed symptom legal escalation handoff with immutable evidence and the smallest safe reproduction.

Primary location	Browser console/network trace, affected record, protected RPC result, Supabase logs, storage path, deployment log, and monitoring event as applicable
Inputs / prerequisites	Timestamp, user/role, record identifier, correlation context, and redacted evidence
Expected outputs	A bounded corrective action, decision record, and verified user-facing outcome
Security or reliability boundary	Do not ask members for passwords, do not expose identity/payment files in support, and do not rewrite history to hide an incident.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Confirm the original symptom is gone, the unauthorized path remains blocked, and a regression test or operational check prevents recurrence.

Scaling note. At scale, use alert thresholds, sampled tracing, dashboards, queue depth, slow-query data, and error budgets to identify the first failing layer.

QA catalogue — Authentication / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the authentication domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Authentication / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the authentication domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Authentication / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the authentication domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Authentication / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the authentication domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Authentication / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the authentication domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Authentication / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the authentication domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Authorization / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the authorization domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Authorization / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the authorization domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Authorization / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the authorization domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Authorization / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the authorization domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Authorization / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the authorization domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Authorization / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the authorization domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Public Posts / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the public posts domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Public Posts / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the public posts domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Public Posts / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the public posts domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Public Posts / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the public posts domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Public Posts / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the public posts domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Public Posts / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the public posts domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Private Media / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the private media domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Private Media / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the private media domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Private Media / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the private media domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Private Media / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the private media domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Private Media / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the private media domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Private Media / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the private media domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Payment Proof / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the payment proof domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Payment Proof / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the payment proof domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Payment Proof / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the payment proof domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Payment Proof / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the payment proof domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Payment Proof / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the payment proof domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Payment Proof / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the payment proof domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Payouts / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the payouts domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Payouts / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the payouts domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Payouts / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the payouts domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Payouts / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the payouts domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Payouts / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the payouts domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Payouts / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the payouts domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Messages / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the messages domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Messages / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the messages domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Messages / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the messages domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Messages / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the messages domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Messages / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the messages domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Messages / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the messages domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Notifications / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the notifications domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Notifications / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the notifications domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Notifications / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the notifications domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Notifications / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the notifications domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Notifications / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the notifications domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Notifications / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the notifications domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Administrator Controls / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the administrator controls domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Administrator Controls / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the administrator controls domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Administrator Controls / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the administrator controls domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Administrator Controls / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the administrator controls domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Administrator Controls / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the administrator controls domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Administrator Controls / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the administrator controls domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Mobile Wrapper / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the mobile wrapper domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Mobile Wrapper / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the mobile wrapper domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Mobile Wrapper / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the mobile wrapper domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Mobile Wrapper / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the mobile wrapper domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Mobile Wrapper / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the mobile wrapper domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Mobile Wrapper / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the mobile wrapper domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Deployment / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the deployment domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Deployment / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the deployment domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Deployment / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the deployment domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Deployment / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the deployment domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Deployment / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the deployment domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Deployment / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the deployment domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Scale Test / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the `scale` test domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Scale Test / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the `scale` test domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Scale Test / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the scale test domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Scale Test / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the `scale` test domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Scale Test / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the `scale` test domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Scale Test / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the `scale` test domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Backup / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the backup domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Backup / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the backup domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Backup / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the backup domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Backup / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the backup domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Backup / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the backup domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Backup / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the backup domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Incident Response / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the incident response domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Incident Response / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the incident response domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Incident Response / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the incident response domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Incident Response / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the incident response domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Incident Response / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the incident response domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Incident Response / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the incident response domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Accessibility / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the accessibility domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Accessibility / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the accessibility domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Accessibility / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the accessibility domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Accessibility / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the accessibility domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Accessibility / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the accessibility domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Accessibility / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the accessibility domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Internationalization / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the internationalization domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Internationalization / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the internationalization domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Internationalization / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the internationalization domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Internationalization / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the internationalization domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Internationalization / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the internationalization domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Internationalization / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the internationalization domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Currency Display / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the currency display domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Currency Display / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the currency display domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Currency Display / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the currency display domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Currency Display / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the currency display domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Currency Display / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the currency display domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Currency Display / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the currency display domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Storage Lifecycle / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the storage lifecycle domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Storage Lifecycle / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the storage lifecycle domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Storage Lifecycle / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the storage lifecycle domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Storage Lifecycle / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the storage lifecycle domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Storage Lifecycle / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the storage lifecycle domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Storage Lifecycle / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the storage lifecycle domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Search / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the search domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Search / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the search domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Search / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the search domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Search / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the search domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Search / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the search domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Search / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the search domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Rate Limits / Happy Path

Repeatable test specification

Purpose. Use this sheet to define a reproducible happy path test for the `rate_limits` domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Rate Limits / Negative Permission

Repeatable test specification

Purpose. Use this sheet to define a reproducible negative permission test for the rate limits domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Rate Limits / Network Interruption

Repeatable test specification

Purpose. Use this sheet to define a reproducible network interruption test for the rate limits domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Rate Limits / Concurrent Update

Repeatable test specification

Purpose. Use this sheet to define a reproducible concurrent update test for the rate limits domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Rate Limits / Large Data Volume

Repeatable test specification

Purpose. Use this sheet to define a reproducible large data volume test for the rate limits domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

QA catalogue — Rate Limits / Release Regression

Repeatable test specification

Purpose. Use this sheet to define a reproducible release regression test for the `rate_limits` domain.

Primary location	Staging environment, seeded non-sensitive accounts, test data identifiers, and relevant feature flag state
Inputs / prerequisites	Explicit preconditions, user role, data state, and expected HTTP/RPC result
Expected outputs	Evidence that the feature works, fails safely, and preserves protected data
Security or reliability boundary	Never use production private documents, payment proof, passwords, or real personal addresses as test fixtures.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. Record result, build/version, responsible reviewer, defects found, and a link to the regression test or incident ticket.

Scaling note. At scale, automate the scenario in CI where practical, run load tests separately, and retain performance baseline results for release comparison.

6 Volume VI — References, Governance, and Living-Document Maintenance

This closing volume establishes update duties. The manual is a governed source: major workflow, storage, payment, privacy, mobile, scaling, or deployment changes require a matching manual update and release record.

Living-manual control — Manual Versioning

Documentation governance sheet

Purpose. Maintain the manual versioning control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control – Chapter Ownership

Documentation governance sheet

Purpose. Maintain the chapter ownership control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Source Map Refresh

Documentation governance sheet

Purpose. Maintain the source map refresh control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control – Diagram Refresh

Documentation governance sheet

Purpose. Maintain the diagram refresh control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Migration Ledger Refresh

Documentation governance sheet

Purpose. Maintain the migration ledger refresh control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Security Review Cadence

Documentation governance sheet

Purpose. Maintain the security review cadence control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Access Review Cadence

Documentation governance sheet

Purpose. Maintain the access review cadence control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Backup Exercise Cadence

Documentation governance sheet

Purpose. Maintain the backup exercise cadence control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Load-Test Cadence

Documentation governance sheet

Purpose. Maintain the load-test cadence control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Incident Drill Cadence

Documentation governance sheet

Purpose. Maintain the incident drill cadence control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Mobile Release Checklist

Documentation governance sheet

Purpose. Maintain the mobile release checklist control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — External Provider Review

Documentation governance sheet

Purpose. Maintain the external provider review control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control – Vendor Outage Communication

Documentation governance sheet

Purpose. Maintain the vendor outage communication control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
Expected outputs	Updated manual section, owner acknowledgment, and next review date
Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Policy Review

Documentation governance sheet

Purpose. Maintain the policy review control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
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Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Data Retention Review

Documentation governance sheet

Purpose. Maintain the data retention review control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
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Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Translation Review

Documentation governance sheet

Purpose. Maintain the translation review control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
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Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control – Accessibility Review

Documentation governance sheet

Purpose. Maintain the accessibility review control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
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Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Cost Review

Documentation governance sheet

Purpose. Maintain the cost review control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
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Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Capacity Review

Documentation governance sheet

Purpose. Maintain the capacity review control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
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Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

Living-manual control — Post-Incident Review

Documentation governance sheet

Purpose. Maintain the post-incident review control so this publication stays aligned with the real platform rather than becoming a historical description.

Primary location	Documentation owner, current release record, change request, affected source/migration, and review date
Inputs / prerequisites	Verified change scope and links to implementation evidence
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Security or reliability boundary	Documentation must never reveal credentials, personal documents, payment proofs, private addresses, or security-bypass instructions.

Problem → Cause → Location → Solution. If this unit fails, first reproduce the smallest safe case. Confirm the caller, authenticated role, record identifier, state value, storage path, and network response in that order. Inspect the named location before editing a dependent screen. Apply a forward-only correction, retain the original evidence, and avoid making an emergency change directly in production.

Verification. A new engineer can locate the source of truth, reproduce the operational workflow, and identify the accountable owner.

Scaling note. At scale, track documentation coverage as an operational quality metric and require updates during change review rather than after an incident.

7 References

- [1] Supabase, **Production Checklist**, <https://supabase.com/docs/guides/deployment/going-into-prod>.
- [2] Render, **Scaling Render Services**, <https://render.com/docs/scaling>.
- [3] Cloudflare, **Cloudflare Cache**, <https://developers.cloudflare.com/cache/>.
- [4] Existing BridgeX repository migration ledger, source tree, tests, native Android shell, and developer operations documents inspected during this edition's generation.

End of Edition 1

This manual is a living technical reference. Update it whenever BridgeX changes a protected workflow, storage policy, database migration, role boundary, mobile release, deployment path, or scaling assumption.